WILDERNESS & URBAN ADVENTURES

This page covers wilderness environments (terrain, hazards, natural obstacles) and the specific challenges of adventuring in cities and towns. See Dungeons & Traps for underground environments, and Weather, Planes & Environmental Rules for climate and planar rules.

WILDERNESS

Outside the safety of city walls, the wilderness is a dangerous place, and many adventurers have gotten lost in its trackless wilds or fallen victim to deadly weather. The following rules give you guidelines on running adventures in a wilderness setting.

GETTING LOST

There are many ways to get lost in the wilderness. Following an obvious road, trail, or feature such as a stream or shoreline prevents most from becoming lost, but travelers striking off cross-country might become disoriented–especially in conditions of poor visibility or in difficult terrain.

Poor Visibility

Anytime characters cannot see at least 60 feet due to reduced visibility conditions, they might become lost. Characters traveling through fog, snow, or a downpour might easily lose the ability to see any landmarks not in their immediate vicinity. Similarly, characters traveling at night might be at risk, too, depending on the quality of their light sources, the amount of moonlight, and whether they have darkvision or low-light vision.

Difficult Terrain

Any character in forest, moor, hill, or mountain terrain might become lost if he moves away from a trail, road, stream, or other obvious path or track. Forests are especially dangerous because they obscure far-off landmarks and make it hard to see the sun or stars.

Chance to Get Lost

If conditions exist that make getting lost a possibility, the character leading the way must succeed on a Survival check or become lost. The difficulty of this check varies based on the terrain, the visibility conditions, and whether or not the character has a map of the area being traveled through. Refer to the table below and use the highest DC that applies.

Terrain

TERRAIN	SURVIVAL DC
Desert or plains	14
Forest	16
Moor or hill	10
Mountain	12
Open sea	18
Urban, ruins, or dungeon	8
Situation	Check Modifier
Proper navigational tools (map, sextant)	+4
Poor visibility	-4
A character with at least 5 ranks in Knowledge (geography) or Knowledge (local) pertaining to the area being traveled through gains a +2 bonus on this check.	

Check once per hour (or portion of an hour) spent in local or overland movement to see if travelers have become lost. In the case of a party moving together, only the character leading the way makes the check.

Effects of Being Lost

If a party becomes lost, it is no longer certain of moving in the direction it intended to travel. Randomly determine the direction in which the party actually travels during each hour of local or overland movement. The characters' movement continues to be random until they blunder into a landmark they can't miss, or until they recognize that they are lost and make an effort to regain their bearings.

Recognizing You're Lost

Once per hour of random travel, each character in the party may attempt a Survival check (DC 20, -1 per hour of random travel) to recognize that he is no longer certain of his direction of travel. Some circumstances might make it obvious that the characters are lost.

Setting a New Course

Determining the correct direction of travel once a party has become lost requires a Survival check (DC 15, +2 per hour of random travel). If a character fails this check, he chooses a random direction as the "correct" direction for resuming travel.

Once the characters are traveling along their new course, correct or incorrect, they might get lost again. If the conditions still make it possible for travelers to become lost, check once per hour of travel as described above to see if the party maintains its new course or begins to move at random again.

Conflicting Directions

It's possible that several characters may attempt to determine the right direction to proceed after becoming lost. Make a Survival check for each character in secret, then tell the players whose characters succeeded the correct direction in which to travel, and tell the players whose characters failed a random direction they think is right, with no indication who is correct.

Regaining Your Bearings

There are several ways for characters to find their way after becoming lost. First, if the characters successfully set a new course and follow it to the destination they're trying to reach, they're not lost anymore. Second, the characters, through random movement, might run into an unmistakable landmark. Third, if conditions suddenly improve–the fog lifts or the sun comes up–lost characters may attempt to set a new course, as described above, with a +4 bonus on the Survival check.

FOREST TERRAIN

Forest terrain can be divided into three categories: sparse, medium, and dense. An immense forest could have all three categories within its borders, with more sparse terrain at the outer edge of the forest and dense forest at its heart.

The table below describes in general terms how likely it is that a given square has a terrain element in it.

Category of Forest

	Category of Forest		
	Sparse	Medium	Dense
Typical trees	50%	70%	80%
Massive trees	—	10%	20%
Light undergrowth	50%	70%	50%
Heavy undergrowth	—	20%	50%

Trees

The most important terrain element in a forest is the trees, obviously. A creature standing in the same square as a tree gains partial cover, which grants a +2 bonus to Armor Class and a +1 bonus on Reflex saves. The presence of a tree doesn't otherwise affect a creature's fighting space, because it's assumed that the creature is using the tree to its advantage when it can. The trunk of a typical tree has AC 4, hardness 5, and 150 hp. A DC 15 Climb check is sufficient to climb a tree. Medium and dense forests have massive trees as well. These trees take up an entire square and provide cover to anyone behind them. They have AC 3, hardness 5, and 600 hp. Like their smaller counterparts, it takes a DC 15 Climb check to climb them.

Undergrowth

Vines, roots, and short bushes cover much of the ground in a forest. A space covered with light undergrowth costs 2 squares of movement to move into, and provides concealment. Undergrowth increases the DC of Acrobatics and Stealth checks by 2 because the leaves and branches get in the way. Heavy undergrowth costs 4 squares of movement to move into and provides concealment with a 30% miss chance (instead of the usual 20%). It increases the DC of Acrobatics checks by 5. Heavy undergrowth is easy to hide in, granting a +5 circumstance bonus on Stealth checks. Running and charging are impossible. Squares with undergrowth are often clustered together. Undergrowth and trees aren't mutually exclusive; it's common for a 5-foot square to have both a tree and undergrowth.

Forest Canopy

It's common for elves and other forest dwellers to live on raised platforms far above the surface floor. These wooden platforms often have rope bridges between them. To get to the treehouses, characters ascend the trees' branches (Climb DC 15), use rope ladders (Climb DC 0), or take pulley elevators (which can be made to rise a number of feet equal to a Strength check, made each round as a full-round action). Creatures on platforms or branches in a forest canopy are considered to have cover when fighting creatures on the ground, and in medium or dense forests they have concealment as well.

Other Forest Terrain Elements

Fallen logs generally stand about 3 feet high and provide cover just as low walls do. They cost 5 feet of movement to cross. Forest streams average 5 to 10 feet wide and no more than 5 feet deep. Pathways wind through most forests, allowing normal movement and providing neither cover nor concealment. These paths are less common in dense forests, but even unexplored forests have occasional game trails.

Stealth and Detection in a Forest

In a sparse forest, the maximum distance at which a Perception check for detecting the nearby presence of others can succeed is 3d6 × 10 feet. In a medium forest, this distance is 2d8 × 10 feet, and in a dense forest it is 2d6 × 10 feet.

Because any square with undergrowth provides concealment, it's usually easy for a creature to use the Stealth skill in the forest. Logs and massive trees provide cover, which also makes hiding possible.

The background noise in the forest makes Perception checks that rely on sound more difficult, increasing the DC of the check by 2 per 10 feet, not 1.

Forest Fires

Most campfire sparks ignite nothing, but if conditions are dry, winds are strong, or the forest floor is dried out and flammable, a forest fire can result. Lightning strikes often set trees ablaze and start forest fires in this way. Whatever the cause of the fire, travelers can get caught in the conflagration.

A forest fire can be spotted from as far away as 2d6 × 100 feet by a character who makes a Perception check, treating the fire as a Colossal creature (reducing the DC by 16). If all characters fail their Perception checks, the fire moves closer to them. They automatically see it when it closes to half the original distance. With proper elevation, the smoke from a forest fire can be spotted as far as 10 miles away.

Characters who are blinded or otherwise unable to make Perception checks can feel the heat of the fire (and thus automatically "spot" it) when it is 100 feet away.

The leading edge of a fire (the downwind side) can advance faster than a human can run (assume 120 feet per round for winds of moderate strength). Once a particular portion of the forest is ablaze, it remains so for 2d4 × 10 minutes before dying to a smoking wasteland. Characters overtaken by a forest fire might find the leading edge of the fire advancing away from them faster than they can keep up, trapping them deeper and deeper within its grasp.

Within the bounds of a forest fire, a character faces three dangers: heat damage, catching on fire, and smoke inhalation.

Heat Damage

Getting caught within a forest fire is even worse than being exposed to extreme heat (see Heat Dangers). Breathing the air causes a character to take 1d6 points of fire damage per round (no save). In addition, a character must make a Fortitude save every 5 rounds (DC 15, +1 per previous check) or take 1d4 points of nonlethal damage. A character who holds his breath can avoid the lethal damage, but not the nonlethal damage. Those wearing heavy clothing or any sort of armor take a -4 penalty on their saving throws. Those wearing metal armor or who come into contact with very hot metal are affected as if by a *heat metal* spell.

Catching on Fire

Characters engulfed in a forest fire are at risk of catching on fire when the leading edge of the fire overtakes them, and continue to be at risk once per minute thereafter.

Smoke Inhalation

Forest fires naturally produce a great deal of smoke. A character who breathes heavy smoke must make a Fortitude save each round (DC 15, +1 per previous check) or spend that round choking and coughing. A character who chokes for 2 consecutive rounds takes 1d6 points of nonlethal damage. Smoke also provides concealment to characters within it.

MARSH TERRAIN

Two categories of marsh exist: relatively dry moors and watery swamps. Both are often bordered by lakes (described in Aquatic Terrain), which are effectively a third category of terrain found in marshes.

Marsh Category

	Marsh Category	
	Moor	Swamp
Shallow bog	20%	40%
Deep bog	5%	20%
Light undergrowth	30%	20%
Heavy undergrowth	10%	20%

Bogs

If a square is part of a shallow bog, it has deep mud or standing water of about 1 foot in depth. It costs 2 squares of movement to move into a square with a shallow bog, and the DC of Acrobatics checks in such a square increases by 2.

A square that is part of a deep bog has roughly 4 feet of standing water. It costs Medium or larger creatures 4 squares of movement to move into a square with a deep bog, or characters can swim if they wish. Small or smaller creatures must swim to move through a deep bog. Tumbling is impossible in a deep bog.

The water in a deep bog provides cover for Medium or larger creatures. Smaller creatures gain improved cover (+8 bonus to AC, +4 bonus on Reflex saves). Medium or larger creatures can crouch as a move action to gain this improved cover. Creatures with this improved cover take a -10 penalty on attacks against creatures that aren't underwater.

Deep bog squares are usually clustered together and surrounded by an irregular ring of shallow bog squares.

Both shallow and deep bogs increase the DC of Stealth checks by 2.

Undergrowth

The bushes, rushes, and other tall grasses in marshes function as undergrowth does in a forest. A square that is part of a bog does not also have undergrowth.

Quicksand

Patches of quicksand present a deceptively solid appearance (appearing as undergrowth or open land) that might trap careless characters. A character approaching a patch of quicksand at a normal pace is entitled to a DC 8 Survival check to spot the danger before stepping in, but charging or running characters don't have a chance to detect a hidden patch before blundering into it. A typical patch of quicksand is 20 feet in diameter; the momentum of a charging or running character carries him 1d2 × 5 feet into the quicksand.

Effects of Quicksand

Characters in quicksand must make a DC 10 Swim check every round to simply tread water in place, or a DC 15 Swim check to move 5 feet in whatever direction is desired. If a trapped character fails this check by 5 or more, he sinks below the surface and begins to drown whenever he can no longer hold his breath (see the Swim skill description in Using Skills).

Characters below the surface of quicksand may swim back to the surface with a successful Swim check (DC 15, +1 per consecutive round of being under the surface).

Rescue

Pulling out a character trapped in quicksand can be difficult. A rescuer needs a branch, spear haft, rope, or similar tool that enables him to reach the victim with one end of it. Then he must make a DC 15 Strength check to successfully pull the victim, and the victim must make a DC 10 Strength check to hold onto the branch, pole, or rope. If both checks succeed, the victim is pulled 5 feet closer to safety. If the victim fails to hold on, he must make a DC 15 Swim check immediately to stay above the surface.

Hedgerows

Common in moors, hedgerows are tangles of stones, soil, and thorny bushes. Narrow hedgerows function as low walls, and it takes 3 squares of movement to cross them. Wide hedgerows are more than 5 feet tall and take up entire squares. They provide total cover, just as a wall does. It takes 4 squares of movement to move through a square with a wide hedgerow; creatures that succeed on a DC 10 Climb check need only 2 squares of movement to move through the square.

Other Marsh Terrain Elements

Some marshes, particularly swamps, have trees just as forests do, usually clustered in small stands. Paths lead across many marshes, winding to avoid bog areas. As in forests, paths allow normal movement and don't provide the concealment that undergrowth does.

Stealth and Detection in a Marsh

In a marsh, the maximum distance at which a Perception check for detecting the nearby presence of others can succeed is 6d6 × 10 feet. In a swamp, this distance is 2d8 × 10 feet.

Undergrowth and deep bogs provide plentiful concealment, so it's easy to use Stealth in a marsh.

HILLS TERRAIN

A hill can exist in most other types of terrain, but hills can also dominate the landscape. Hills terrain is divided into two categories: gentle hills and rugged hills. Hills terrain often serves as a transition zone between rugged terrain such as mountains and flat terrain such as plains.

Hills Category

Hills Category	Gentle Hills	Rugged Hills
Gradual slope	75%	40%
Steep slope	20%	50%
Cliff	5%	10%
Light undergrowth	15%	15%

Gradual Slope
This incline isn't steep enough to affect movement, but characters gain a +1 bonus on melee attacks against foes downhill from them.

Steep Slope
Characters moving uphill (to an adjacent square of higher elevation) must spend 2 squares of movement to enter each square of steep slope. Characters running or charging downhill (moving to an adjacent square of lower elevation) must succeed on a DC 10 Acrobatics check upon entering the first steep slope square. Mounted characters make a DC 10 Ride check instead. Characters who fail this check stumble and must end their movement 1d2 × 5 feet later. Characters who fail by 5 or more fall prone in the square where they end their movement. A steep slope increases the DC of Acrobatics checks by 2.

Cliff
A cliff typically requires a DC 15 Climb check to scale and is 1d4 × 10 feet tall, although the needs of your map might mandate a taller cliff. A cliff isn't perfectly vertical, taking up 5-foot squares if it's less than 30 feet tall and 10-foot squares if it's 30 feet or taller.

Light Undergrowth
Sagebrush and other scrubby bushes grow on hills, although they rarely cover the landscape. Light undergrowth provides concealment and increases the DC of Acrobatics and Stealth checks by 2.

Other Hills Terrain Elements
Trees aren't out of place in hills terrain, and valleys often have active streams (5 to 10 feet wide and no more than 5 feet deep) or dry streambeds (treat as a trench 5 to 10 feet across) in them. If you add a stream or streambed, remember that water always flows downhill.

Stealth and Detection in Hills

In gentle hills, the maximum distance at which a Perception check for detecting the nearby presence of others can succeed is 2d10 × 10 feet. In rugged hills, this distance is 2d6 × 10 feet.

Hiding in hills terrain can be difficult if there isn't undergrowth around. A hilltop or ridge provides enough cover to hide from anyone below the hilltop or ridge.

MOUNTAIN TERRAIN

The three mountain terrain categories are alpine meadows, rugged mountains, and forbidding mountains. As characters ascend into a mountainous area, they're likely to face each terrain category in turn, beginning with alpine meadows, extending through rugged mountains, and reaching forbidding mountains near the summit.

Mountains have an important terrain element, the rock wall, that is marked on the border between squares rather than taking up squares itself.

Mountain Category

	Mountain Category		
	Alpine Meadow	Rugged	Forbidding
Gradual slope	50%	25%	15%
Steep slope	40%	55%	55%
Cliff	10%	15%	20%
Chasm	—	5%	10%
Light undergrowth	20%	10%	–
Scree	—	20%	30%
Dense rubble	—	20%	30%

Gradual and Steep Slopes

These function as described in Hills Terrain.

Cliff

These terrain elements also function like their hills terrain counterparts, but they're typically 2d6 × 10 feet tall. Cliffs taller than 80 feet take up 20 feet of horizontal space.

Chasm

Usually formed by natural geological processes, chasms function like pits in a dungeon setting. Chasms aren't hidden, so characters won't fall into them by accident (although bull rushes are another story). A typical chasm is 2d4 × 10 feet deep, at least 20 feet long, and anywhere from 5 feet to 20 feet wide. It takes a DC 15 Climb check to climb out of a chasm. In forbidding mountain terrain, chasms are typically 2d8 × 10 feet deep.

Light Undergrowth

This functions as described in Forest Terrain.

Scree
A field of shifting gravel, scree doesn't affect speed, but it can be treacherous on a slope. The DC of Acrobatics checks increases by 2 if there's scree on a gradual slope and by 5 if there's scree on a steep slope. The DC of Stealth checks increases by 2 if the scree is on a slope of any kind.

Dense Rubble
The ground is covered with rocks of all sizes. It costs 2 squares of movement to enter a square with dense rubble. The DC of Acrobatics checks on dense rubble increases by 5, and the DC of Stealth checks increases by 2.

Rock Wall
A vertical plane of stone, rock walls require DC 25 Climb checks to ascend. A typical rock wall is 2d4 × 10 feet tall in rugged mountains and 2d8 × 10 feet tall in forbidding mountains. Rock walls are drawn on the edges of squares, not in the squares themselves.

Cave Entrance
Found in cliff and steep slope squares and next to rock walls, cave entrances are typically between 5 and 20 feet wide and 5 feet deep. A cave could be anything from a simple chamber to the entrance to an elaborate dungeon. Caves used as monster lairs typically have 1d3 rooms that are 1d4 × 10 feet across.

Other Mountain Terrain Features
Most alpine meadows begin above the treeline, so trees and other forest elements are rare in the mountains. Mountain terrain can include active streams (5 to 10 feet wide and no more than 5 feet deep) and dry streambeds (treat as a trench 5 to 10 feet across). Particularly high-altitude areas tend to be colder than the lowland areas that surround them, so they might be covered in ice sheets (described in Desert Terrain).

Stealth and Detection in Mountains
As a guideline, the maximum distance in mountain terrain at which a Perception check for detecting the nearby presence of others can succeed is 4d10 × 10 feet. Certain peaks and ridgelines afford much better vantage points, of course, and twisting valleys and canyons have much shorter spotting distances. Because there's little vegetation to obstruct line of sight, the specifics on your map are your best guide for the range at which an encounter could begin. As in hills terrain, a ridge or peak provides enough cover to hide from anyone below the high point.

It's easier to hear faraway sounds in the mountains. The DC of Perception checks that rely on sound increase by 1 per 20 feet between listener and source, not per 10 feet.

Avalanches

The combination of high peaks and heavy snowfalls means that avalanches are a deadly peril in many mountainous areas. While avalanches of snow and ice are common, it's also possible to have an avalanche of rock and soil.

An avalanche can be spotted from as far away as 1d10 × 500 feet by a character who makes a DC 20 Perception check, treating the avalanche as a Colossal creature. If all characters fail their Perception checks to determine the encounter distance, the avalanche moves closer to them, and they automatically become aware of it when it closes to half the original distance. It's possible to hear an avalanche coming even if you can't see it. Under optimum conditions (no other loud noises occurring), a character who makes a DC 15 Perception check can hear the avalanche or landslide when it is 1d6 × 500 feet away. This check might have a DC of 20, 25, or higher in conditions where hearing is difficult (such as in the middle of a thunderstorm).

A landslide or avalanche consists of two distinct areas: the bury zone (in the direct path of the falling debris) and the slide zone (the area the debris spreads out to encompass). Characters in the bury zone always take damage from the avalanche; characters in the slide zone might be able to get out of the way. Characters in the bury zone take 8d6 points of damage, or half that amount if they make a DC 15 Reflex save. They are subsequently buried. Characters in the slide zone take 3d6 points of damage, or no damage if they make a DC 15 Reflex save. Those who fail their saves are buried.

Buried characters take 1d6 points of nonlethal damage per minute. If a buried character falls unconscious, he must make a DC 15 Constitution check or take 1d6 points of lethal damage each minute thereafter until freed or dead. See Cave-Ins and Collapses for rules on digging out buried creatures.

The typical avalanche has a width of 1d6 × 100 feet, from one edge of the slide zone to the opposite edge. The bury zone in the center of the avalanche is half as wide as the avalanche's full width.

To determine the precise location of characters in the path of an avalanche, roll 1d6 × 20; the result is the number of feet from the center of the path taken by the bury zone to the center of the party's location. Avalanches of snow and ice advance at a speed of 500 feet per round, while rock and soil avalanches travel at a speed of 250 feet per round.

Mountain Travel

High altitude travel can be extremely fatiguing–and sometimes deadly–to creatures that aren't used to it. Cold becomes extreme, and the lack of oxygen in the air can wear down even the most hardy of warriors.

Acclimated Characters

Creatures accustomed to high altitude generally fare better than lowlanders. Any creature with an Environment entry that includes mountains is considered native to the area and acclimated to the high altitude. Characters can also acclimate themselves by living at high altitude for a month. Characters who spend more than two months away from the mountains must reacclimate themselves when they return. Undead, constructs, and other creatures that do not breathe are immune to altitude effects.

Altitude Zones

In general, mountains present three possible altitude bands: low pass, low peak/high pass, and high peak.

Low Pass (lower than 5,000 feet)

Most travel in low mountains takes place in low passes, a zone consisting largely of alpine meadows and forests. Travelers might find the going difficult (which is reflected in the movement modifiers for traveling through mountains), but the altitude itself has no game effect.

Low Peak or High Pass (5,000 to 15,000 feet)

Ascending to the highest slopes of low mountains, or most normal travel through high mountains, falls into this category. All non-acclimated creatures labor to breathe in the thin air at this altitude. Characters must succeed on a Fortitude save each hour (DC 15, +1 per previous check) or be fatigued. The fatigue ends when the character descends to an altitude with more air. Acclimated characters do not have to attempt the Fortitude save.

High Peak (more than 15,000 feet)

The highest mountains exceed 15,000 feet in height. At these elevations, creatures are subject to both high altitude fatigue (as described above) and altitude sickness, whether or not they're acclimated to high altitudes. Altitude sickness represents long-term oxygen deprivation, and affects mental and physical ability scores. After each 6-hour period a character spends at an altitude of over 15,000 feet, he must succeed on a Fortitude save (DC 15, +1 per previous check) or take 1 point of damage to all ability scores. Creatures acclimated to high altitude receive a +4 competence bonus on their saving throws to resist high altitude effects and altitude sickness, but eventually even seasoned mountaineers must abandon these dangerous elevations.

DESERT TERRAIN

Desert terrain exists in warm, temperate, and cold climates, but all deserts share one common trait: little rain.
The three categories of desert terrain are tundra (cold desert), rocky deserts (often temperate), and sandy deserts (often warm).

Tundra differs from the other desert categories in two important ways. Because snow and ice cover much of the landscape, it's easy to find water. During the height of summer, the permafrost thaws to a depth of a foot or so, turning the landscape into a vast field of mud. The muddy tundra affects movement and skill use as the shallow bogs described in Marsh Terrain, although there's little standing water.

The table below describes terrain elements found in each of the three desert categories. The terrain elements on this table are mutually exclusive; for instance, a square of tundra might contain either light undergrowth or an ice sheet, but not both.

Desert Category

	Desert Category		
	Tundra	Rocky	Sandy
Light undergrowth	15%	5%	5%
Ice sheet	25%	—	–
Light rubble	5%	30%	10%
Dense rubble	—	30%	5%
Sand dunes	—	—	50%

Light Undergrowth

Consisting of scrubby, hardy bushes and cacti, light undergrowth functions as described for other terrain types.

Ice Sheet
The ground is covered with slippery ice. It costs 2 squares of movement to enter a square covered by an ice sheet, and the DC of Acrobatics checks there increases by 5. A DC 10 Acrobatics check is required to run or charge across an ice sheet.

Light Rubble
Small rocks are strewn across the ground, making nimble movement more difficult. The DC of Acrobatics checks increases by 2.

Dense Rubble
This terrain feature consists of more and larger stones. It costs 2 squares of movement to enter a square with dense rubble. The DC of Acrobatics checks increases by 5, and the DC of Stealth checks increases by 2.

Sand Dunes
Created by the action of wind on sand, dunes function as hills that move. If the wind is strong and consistent, a sand dune can move several hundred feet in a week's time. Sand dunes can cover hundreds of squares. They always have a gentle slope pointing in the direction of the prevailing wind and a steep slope on the leeward side.

Other Desert Terrain Features
Tundra is sometimes bordered by forests, and the occasional tree isn't out of place in the cold wastes. Rocky deserts have towers and mesas consisting of flat ground surrounded on all sides by cliffs and steep slopes (as described in Mountain Terrain). Sandy deserts sometimes have quicksand; this functions as described in Marsh Terrain, although desert quicksand is a waterless mixture of fine sand and dust. All desert terrain is crisscrossed with dry streambeds (treat as trenches 5 to 15 feet wide) that fill with water on the rare occasions when rain falls.

Stealth and Detection in the Desert
In general, the maximum distance in desert terrain at which a Perception check for detecting the nearby presence of others can succeed is 6d6 × 20 feet; beyond this distance, elevation changes and heat distortion in warm deserts makes sight-based Perception impossible. The presence of dunes in sandy deserts limits spotting distance to 6d6 × 10 feet. The scarcity of undergrowth or other elements that offer concealment or cover makes using Stealth more difficult.

Sandstorms
A sandstorm reduces visibility to 1d10 × 5 feet and provides a -4 penalty on Perception checks. A sandstorm deals 1d3 points of nonlethal damage per hour to any creatures caught in the open, and leaves a thin coating of sand in its wake. Driving sand creeps in through all but the most secure seals and seams, chafing skin and contaminating carried gear.

PLAINS TERRAIN

Plains come in three categories: farms, grasslands, and battlefields. Farms are common in settled areas, while grasslands represent untamed plains. The battlefields where large armies clash are temporary places, usually reclaimed by natural vegetation or the farmer's plow. Battlefields represent a third terrain category because adventurers tend to spend a lot of time there, not because they're particularly prevalent.

The table below shows the proportions of terrain elements in the different categories of plains. On a farm, light undergrowth represents most mature grain crops, so farms growing vegetable crops will have less light undergrowth, as will all farms during the time between harvest and a few months after planting.

The terrain elements in the table below are mutually exclusive.

Plains Category

	Plains Category		
	Farm	Grassland	Battlefield
Light undergrowth	40%	20%	10%
Heavy undergrowth	—	10%	–
Light rubble	—	—	10%
Trench	5%	—	5%
Berm	—	—	5%

Undergrowth

Whether they're crops or natural vegetation, the tall grasses of the plains function like light undergrowth in a forest. Particularly thick bushes form patches of heavy undergrowth that dot the landscape in grasslands.

Light Rubble

On the battlefield, light rubble usually represents something that was destroyed: the ruins of a building or the scattered remnants of a stone wall, for example. It functions as described in the Desert Terrain section.

Trench

Often dug before a battle to protect soldiers, a trench functions as a low wall, except that it provides no cover against adjacent foes. It costs 2 squares of movement to leave a trench, but it costs nothing extra to enter one. Creatures outside a trench who make a melee attack against a creature inside the trench gain a +1 bonus on melee attacks because they have higher ground. In farm terrain, trenches are generally irrigation ditches.

Berm

A common defensive structure, a berm is a low, earthen wall that slows movement and provides a measure of cover. Put a berm on the map by drawing two adjacent rows of steep slope (described in Hills Terrain), with the edges of the berm on the downhill side. Thus, a character crossing a 2-square berm will travel uphill for 1 square, then downhill for 1 square. Two square berms provide cover as low walls for anyone standing behind them. Larger berms provide the low wall benefit for anyone standing 1 square downhill from the top of the berm.

Fences

Wooden fences are generally used to contain livestock or impede oncoming soldiers. It costs an extra square of movement to cross a wooden fence. A stone fence provides a measure of cover as well, functioning as low walls. Mounted characters can cross a fence without slowing their movement if they succeed on a DC 15 Ride check. If the check fails, the steed crosses the fence, but the rider falls out of the saddle.

Other Plains Terrain Features

Occasional trees dot the landscape in many plains, although on battlefields they're often felled to provide raw material for siege engines (described in Urban Features). Hedgerows (described in Marsh Terrain) are found in plains as well. Streams, generally 5 to 20 feet wide and 5 to 10 feet deep, are commonplace.

Stealth and Detection in Plains

In plains terrain, the maximum distance at which a Perception check for detecting the nearby presence of others can succeed is 6d6 × 40 feet, although the specifics of your map might restrict line of sight. Cover and concealment are not uncommon, so a good place of refuge is often nearby, if not right at hand.

AQUATIC TERRAIN

Aquatic terrain is the least hospitable to most PCs, because they can't breathe there. Aquatic terrain doesn't offer the variety that land terrain does. The ocean floor holds many marvels, including undersea analogues of any of the terrain elements described earlier in this section, but if characters find themselves in the water because they were bull rushed off the deck of a pirate ship, the tall kelp beds hundreds of feet below them don't matter. Accordingly, these rules simply divide aquatic terrain into two categories: flowing water (such as streams and rivers) and non-flowing water (such as lakes and oceans).

Flowing Water

Large, placid rivers move at only a few miles per hour, so they function as still water for most purposes. But some rivers and streams are swifter; anything floating in them moves downstream at a speed of 10 to 40 feet per round. The fastest rapids send swimmers bobbing downstream at 60 to 90 feet per round. Fast rivers are always at least rough water (Swim DC 15), and whitewater rapids are stormy water (Swim DC 20). If a character is in moving water, move her downstream the indicated distance at the end of her turn. A character trying to maintain her position relative to the riverbank can spend some or all of her turn swimming upstream.

Swept Away

Characters swept away by a river moving 60 feet per round or faster must make DC 20 Swim checks every round to avoid going under. If a character gets a check result of 5 or more over the minimum necessary, she arrests her motion by catching a rock, tree limb, or bottom snag–she is no longer being carried along by the flow of the water. Escaping the rapids by reaching the bank requires three DC 20 Swim checks in a row. Characters arrested by a rock, limb, or snag can't escape under their own power unless they strike out into the water and attempt to swim their way clear. Other characters can rescue them as if they were trapped in quicksand (described in Marsh Terrain).

Non-Flowing Water

Lakes and oceans simply require a swim speed or successful Swim checks to move through (DC 10 in calm water, DC 15 in rough water, DC 20 in stormy water). Characters need a way to breathe if they're underwater; failing that, they risk drowning. When underwater, characters can move in any direction.

Stealth and Detection Underwater

How far you can see underwater depends on the water's clarity. As a guideline, creatures can see 4d8 × 10 feet if the water is clear, and 1d8 × 10 feet if it's murky. Moving water is always murky, unless it's in a particularly large, slow-moving river.

It's hard to find cover or concealment to hide underwater (except along the sea floor).

Invisibility

An invisible creature displaces water and leaves a visible, body-shaped "bubble" where the water was displaced. The creature still has concealment (20% miss chance), but not total concealment (50% miss chance).

Underwater Combat

Land-based creatures can have considerable difficulty when fighting in water. Water affects a creature's attack rolls, damage, and movement. In some cases a creature's opponents might get a bonus on attacks. The effects are summarized on Table: Combat Adjustments Underwater. They apply whenever a character is swimming, walking in chest-deep water, or walking along the bottom of a body of water.

Table: Combat Adjustments Underwater

CONDITION	ATTACK/DAMAGE	MOVEMENT	OFF BALANCE?1
Slashing or Bludgeoning	Piercing		
Freedom of movement	normal/normal	normal/normal	normal
Has a swim speed	-2/half	normal	normal
Successful Swim check	-2/half2	normal	quarter or half3
Firm footing4	-2/half2	normal	half
None of the above	-2/half2	-2/half	normal
1 Creatures flailing about in the water (usually because they failed their Swim checks) have a hard time fighting effectively. An off-balance creature loses its Dexterity bonus to Armor Class, and opponents gain a +2 bonus on attacks against it.			
2 A creature without *freedom of movement* effects or a swim speed makes grapple checks underwater at a -2 penalty, but deals damage normally when grappling.			
3 A successful Swim check lets a creature move one-quarter its speed as a move action or one-half its speed as a full-round action.			
4 Creatures have firm footing when walking along the bottom, braced against a ship's hull, or the like. A creature can only walk along the bottom if it wears or carries enough gear to weigh itself down: at least 16 pounds for Medium creatures, twice that for each size category larger than Medium, and half that for each size category smaller than Medium.			

Ranged Attacks Underwater

Thrown weapons are ineffective underwater, even when launched from land. Attacks with other ranged weapons take a -2 penalty on attack rolls for every 5 feet of water they pass through, in addition to the normal penalties for range.

Attacks from Land

Characters swimming, floating, or treading water on the surface, or wading in water at least chest deep, have improved cover (+8 bonus to AC, +4 bonus on Reflex saves) from opponents on land. Land-bound opponents who have *freedom of movement* effects ignore this cover when making melee attacks against targets in the water. A completely submerged creature has total cover against opponents on land unless those opponents have *freedom of movement* effects. Magical effects are unaffected except for those that require attack rolls (which are treated like any other effects) and fire effects.

Fire

Nonmagical fire (including alchemist's fire) does not burn underwater. Spells or spell-like effects with the fire descriptor are ineffective underwater unless the caster makes a caster level check (DC 20 + spell level). If the check succeeds, the spell creates a bubble of steam instead of its usual fiery effect, but otherwise the spell works as described. A supernatural fire effect is ineffective underwater unless its description states otherwise. The surface of a body of water blocks line of effect for any fire spell. If the caster has made the caster level check to make the fire spell usable underwater, the surface still blocks the spell's line of effect.

Spellcasting Underwater

Casting spells while submerged can be difficult for those who cannot breathe underwater. A creature that cannot breathe water must make a concentration check (DC 15 + spell level) to cast a spell underwater (this is in addition to the caster level check to successfully cast a fire spell underwater). Creatures that can breathe water are unaffected and can cast spells normally. Some spells might function differently underwater, subject to GM discretion.

Floods

In many wilderness areas, river floods are a common occurrence.

In spring, an enormous snowmelt can engorge the streams and rivers it feeds. Other catastrophic events such as massive rainstorms or the destruction of a dam can create floods as well.

During a flood, rivers become wider, deeper, and swifter. Assume that a river rises by 1d10+10 feet during the spring flood, and its width increases by a factor of 1d4 × 50%. Fords might disappear for days, bridges might be swept away, and even ferries might not be able to manage the crossing of a flooded river. A river in flood makes Swim checks one category harder (calm water becomes rough, and rough water becomes stormy). Rivers also become 50% swifter.

URBAN ADVENTURES

At first glance, a city is much like a dungeon, made up of walls, doors, rooms, and corridors. Adventures that take place in cities have two salient differences from their dungeon counterparts, however. Characters have greater access to resources, and they must contend with law enforcement.

ACCESS TO RESOURCES

Unlike in dungeons and the wilderness, characters can buy and sell gear quickly in a city. A large city or metropolis probably has high-level NPCs and experts in obscure fields of knowledge who can provide assistance and decipher clues. And when the PCs are battered and bruised, they can retreat to the comfort of a room at an inn.

The freedom to retreat and ready access to the marketplace means that the players have a greater degree of control over the pacing of an urban adventure.

LAW ENFORCEMENT

The other key distinctions between adventuring in a city and delving into a dungeon is that a dungeon is, almost by definition, a lawless place where the only law is that of the jungle: kill or be killed. A city, on the other hand, is held together by a code of laws, many of which are explicitly designed to prevent the sort of killing and looting that adventurers engage in all the time. Even so, most cities' laws recognize monsters as a threat to the stability the city relies on, and prohibitions about murder rarely apply to monsters such as aberrations or evil outsiders. Most evil humanoids, however, are typically protected by the same laws that protect all the citizens of the city. Having an evil alignment is not a crime (except in some severely theocratic cities, perhaps, with the magical power to back up the law); only evil deeds are against the law. Even when adventurers encounter an evildoer in the act of perpetrating some heinous evil upon the populace of the city, the law tends to frown on the sort of vigilante justice that leaves the evildoer dead or otherwise unable to testify at a trial.

WEAPON AND SPELL RESTRICTIONS

Different cities have different laws about such issues as carrying weapons in public and restricting spellcasters.

The city's laws might not affect all characters equally. A monk isn't hampered at all by a law about peace-bonding weapons, but a cleric is reduced to a fraction of his power if all holy symbols are confiscated at the city's gates.

URBAN FEATURES

Walls, doors, poor lighting, and uneven footing: in many ways a city is much like a dungeon. Some special considerations for an urban setting are covered below.

Walls and Gates

Many cities are surrounded by walls. A typical small city wall is a fortified stone wall 5 feet thick and 20 feet high. Such a wall is fairly smooth, requiring a DC 30 Climb check to scale. The walls are crenellated on one side to provide a low wall for the guards atop it, and there is just barely room for guards to walk along the top of the wall. A typical small city wall has AC 3, hardness 8, and 450 hp per 10-foot section.

A typical large city wall is 10 feet thick and 30 feet high, with crenellations on both sides for the guards on top of the wall. It is likewise smooth, requiring a DC 30 Climb check to scale. Such a wall has AC 3, hardness 8, and 720 hp per 10-foot section.

A typical metropolis wall is 15 feet thick and 40 feet tall. It has crenellations on both sides and often has a tunnel and small rooms running through its interior. Metropolis walls have AC 3, hardness 8, and 1,170 hp per 10-foot section.

Unlike smaller cities, metropolises often have interior walls as well as surrounding walls–either old walls that the city has outgrown, or walls dividing individual districts from each other. Sometimes these walls are as large and thick as the outer walls, but more often they have the characteristics of a large city's or small city's walls.

Watchtowers

Some city walls are adorned with watchtowers set at irregular intervals. Few cities have enough guards to keep someone constantly stationed at every tower, unless the city is expecting attack from outside. The towers provide a superior view of the surrounding countryside as well as a point of defense against invaders.

Watchtowers are typically 10 feet higher than the wall they adjoin, and their diameter is 5 times the thickness of the wall. Arrow slits line the outer sides of the upper stories of a tower, and the top is crenellated like the surrounding walls are. In a small tower (25 feet in diameter adjoining a 5-foot-thick wall), a simple ladder typically connects the tower's stories and its roof. In a larger tower, stairs serve that purpose.

Heavy wooden doors, reinforced with iron and bearing good locks (Disable Device DC 30), block entry to a tower, unless the tower is in regular use. As a rule, the captain of the guard keeps the keys to the towers secured on her person, and second copies are in the city's inner fortress or barracks.

Gates

A typical city gate is a gatehouse with two portcullises and murder holes above the space between them. In towns and some small cities, the primary entry is through iron double doors set into the city wall.

Gates are usually open during the day and locked or barred at night. Usually, one gate lets in travelers after sunset and is staffed by guards who will open it for someone who seems honest, presents proper papers, or offers a large enough bribe (depending on the city and the guards).

Guards and Soldiers

A city typically has full-time military personnel equal to 1% of its adult population, in addition to militia or conscript soldiers equal to 5% of the population. The full-time soldiers are city guards responsible for maintaining order within the city, similar to the role of modern police, and (to a lesser extent) for defending the city from outside assault. Conscript soldiers are called up to serve in case of an attack on the city.

A typical city guard force works on three 8-hour shifts, with 30% of the force on a day shift (8 a.m. to 4 p.m.), 35% on an evening shift (4 p.m. to 12 a.m.), and 35% on a night shift (12 a.m. to 8 a.m.). At any given time, 80% of the guards on duty are on the streets patrolling, while the remaining 20% are stationed at various posts throughout the city where they can respond to nearby alarms. At least one such guard post is present within each neighborhood of a city (each neighborhood consisting of several districts).

The majority of a city guard force is made up of warriors, mostly 1st level. Officers include higher-level warriors, fighters, a fair number of clerics, and wizards or sorcerers, as well as multiclass fighter/spellcasters.

Siege Engines

Siege engines are large weapons, temporary structures, or pieces of equipment traditionally used in besieging castles or fortresses.

Table: Siege Engines

ITEM	COST	DAMAGE	CRITICAL RANGE	INCREMENT	TYPICAL CREW
Catapult, heavy	800 gp	6d6	–	200 ft. (100 ft. minimum)	4
Catapult, light	550 gp	4d6	–	150 ft. (100 ft. minimum)	2
Ballista	500 gp	3d8	19-20	120 ft.	1
Ram	1,000 gp	3d6*	–	–	10
Siege tower	2,000 gp	–	–	–	20

- See description for special rules.

Catapult Attack Modifiers

CONDITION	MODIFIER
No line of sight to target square	-6
Successive shots (crew can see where most recent misses landed)	Cumulative +2 per previous miss (maximum +10)
Successive shots (crew can't see where most recent misses landed, but observer is providing feedback)	Cumulative +1 per previous miss (maximum +5)
Siege engines are treated as difficult devices if someone tries to disable them using Disable Device. This takes 2d4 rounds and requires a DC 20 Disable Device check. Siege engines are typically made out of wood and have an AC of 3 (-5 Dex, -2 size), a Hardness of 5, and 80 hit points. Siege engines made up of a different material might have different values. Some siege engines are armored as well. Treat the siege engine as a Huge creature to determine the cost of such armor. Siege engines can be crafted as masterwork and enchanted as magic weapons, adding bonuses on attack rolls to the checks made to hit with the siege engine. A masterwork siege engine costs 300 gp more than the listed price. Enchanting a siege engine costs twice the normal amount. For example, a *+1 flaming heavy catapult* , armored with full plate, would have an AC of 11 and would cost 23,100 gp (800 gp base + 6,000 gp for the armor + 300 gp masterwork + 16,000 gp for the enhancements).	

Catapult, Heavy

A heavy catapult is a massive engine capable of throwing rocks or heavy objects with great force. Because the catapult throws its payload in a high arc, it can hit squares out of its line of sight. To fire a heavy catapult, the crew chief makes a special check against DC 15 using only his base attack bonus, Intelligence modifier, range increment penalty, and the appropriate modifiers from the lower section of Table: Siege Engines. If the check succeeds, the catapult stone hits the square the catapult was aimed at, dealing the indicated damage to any object or character in the square. Characters who succeed on a DC 15 Reflex save take half damage. Once a catapult stone hits a square, subsequent shots hit the same square unless the catapult is reaimed or the wind changes direction or speed.

If a catapult stone misses, roll 1d8 to determine where it lands. This determines the misdirection of the throw, with 1 being back toward the catapult and 2 through 8 counting clockwise around the target square. Finally, count 1d4 squares away from the target square for every range increment of the attack.

Loading a catapult requires a series of full-round actions. It takes a DC 15 Strength check to winch the throwing arm down; most catapults have wheels to allow up to two crew members to use the aid another action, assisting the main winch operator. A DC 15 Profession (siege engineer) check latches the arm into place, and then another DC 15 Profession (siege engineer) check loads the catapult ammunition. It takes four full-round actions to reaim a heavy catapult (multiple crew members can perform these full-round actions in the same round, so it would take a crew of four only 1 round to reaim the catapult).

A heavy catapult takes up a space 15 feet across.

Catapult, Light

This is a smaller, lighter version of the heavy catapult. It functions as the heavy catapult, except that it takes a DC 10 Strength check to winch the arm into place, and only two full-round actions are required to reaim the catapult.

A light catapult takes up a space 10 feet across.

Ballista

A ballista is essentially a Huge heavy crossbow fixed in place. Its size makes it hard for most creatures to aim it. Thus, a Medium creature takes a -4 penalty on attack rolls when using a ballista, and a Small creature takes a -6 penalty. It takes a creature smaller than Large two full-round actions to reload the ballista after firing.

A ballista takes up a space 5 feet across.

Ram

This heavy pole is sometimes suspended from a movable scaffold that allows the crew to swing it back and forth against objects. As a full-round action, the character closest to the front of the ram makes an attack roll against the AC of the construction, applying the -4 penalty for lack of proficiency. It's not possible to be proficient with this device. In addition to the damage given on Table: Siege Engines, up to nine other characters holding the ram can add their Strength modifiers to the ram's damage, if they devote an attack action to doing so. It takes at least one Huge or larger creature, two Large creatures, four Medium creatures, or eight Small creatures to swing a ram.

A ram is typically 30 feet long. In a battle, the creatures wielding the ram stand in two adjacent columns of equal length, with the ram between them.

Siege Tower

This device is a massive wooden tower on wheels or rollers that can be rolled up against a wall to allow attackers to scale the tower and thus get to the top of the wall with cover. The wooden walls are usually 1 foot thick.

A typical siege tower takes up a space 15 feet across. The creatures inside push it at a base land speed of 10 feet (and a siege tower can't run). The eight creatures pushing on the ground floor have total cover, and those on higher floors get improved cover and can fire through arrow slits.

City Streets

Typical city streets are narrow and twisting. Most streets average 15 to 20 feet wide, while alleys range from 10 feet wide to only 5 feet. Cobblestones in good condition allow normal movement, but roads in poor repair and heavily rutted dirt streets are considered light rubble, increasing the DC of Acrobatics checks by 2.

Some cities have no larger thoroughfares, particularly cities that gradually grew from small settlements to larger cities. Cities that are planned, or perhaps have suffered a major fire that allowed authorities to construct new roads through formerly inhabited areas, might have a few larger streets through town. These main roads are 25 feet wide–offering room for wagons to pass each other–with 5-foot-wide sidewalks on either side.

Crowds

Urban streets are often full of people going about their daily lives. In most cases, it isn't necessary to put every 1st-level commoner on the map when a fight breaks out on the city's main thoroughfare. Instead, just indicate which squares on the map contain crowds. If crowds see something obviously dangerous, they'll move away at 30 feet per round at initiative count 0. It takes 2 squares of movement to enter a square with crowds. The crowds provide cover for anyone who does so, enabling a Stealth check and providing a bonus to Armor Class and on Reflex saves.

Directing Crowds

It takes a DC 15 Diplomacy check or DC 20 Intimidate check to convince a crowd to move in a particular direction, and the crowd must be able to hear or see the character making the attempt. It takes a full-round action to make the Diplomacy check, but only a free action to make the Intimidate check.

If two or more characters are trying to direct a crowd in different directions, they make opposed Diplomacy or Intimidate checks to determine to whom the crowd listens. The crowd ignores everyone if none of the characters' check results beat the DCs given above.

Above and Beneath the Streets

Rooftops

Getting to a roof usually requires climbing a wall, unless the character can reach a roof by jumping down from a higher window, balcony, or bridge. Flat roofs, common only in warm climates (as accumulated snow can cause a flat roof to collapse), are easy to run across. Moving along the peak of a pitched roof requires a DC 20 Acrobatics check. Moving on an angled roof surface without changing altitude (moving parallel to the peak, in other words) requires a DC 15 Acrobatics check. Moving up and down across the peak of a roof requires a DC 10 Acrobatics check.

Eventually a character runs out of roof, requiring a long jump across to the next roof or down to the ground. The distance to the closest roof is usually 1d3 × 5 feet horizontally, but the next roof is equally likely to be 5 feet higher, 5 feet lower, or the same height. Use the guidelines in the Acrobatics skill (a horizontal jump's peak height is one-fourth of the horizontal distance) to determine whether a character can make a jump.

Sewers

To get into the sewers, most characters open a grate (a full-round action) and jump down 10 feet. Sewers are built exactly like dungeons, except that they're much more likely to have floors that are slippery or covered with water. Sewers are also similar to dungeons in terms of creatures liable to be encountered therein. Some cities were built atop the ruins of older civilizations, so their sewers sometimes lead to treasures and dangers from a bygone age.

City Buildings

Most city buildings fall into three categories. The majority of buildings in the city are two to five stories high, built side-by-side to form long rows separated by secondary or main streets. These row houses usually have businesses on the ground floor, with offices or apartments above.

Inns, successful businesses, and large warehouses–as well as millers, tanners, and other businesses that require extra space–are generally large, free-standing buildings with up to five stories.

Finally, small residences, shops, warehouses, or storage sheds are simple, one-story wooden buildings, especially if they're in poorer neighborhoods.

Most city buildings are made of a combination of stone or clay brick (on the lower one or two stories) and timbers (for the upper stories, interior walls, and floors). Roofs are a mixture of boards, thatch, and slates, sealed with pitch. A typical lower-story wall is 1 foot thick, with AC 3, hardness 8, 90 hp, and a Climb DC of 25. Upper-story walls are 6 inches thick, with AC 3, hardness 5, 60 hp, and a Climb DC of 21. Exterior doors on most buildings are good wooden doors that are usually kept locked, except on public buildings such as shops and taverns.

City Lights

If a city has main thoroughfares, they are lined with lanterns hanging at a height of 7 feet from building awnings. These lanterns are spaced 60 feet apart, so their illumination is all but continuous. Secondary streets and alleys are not lit; it is common for citizens to hire lantern-bearers when going out after dark.

Alleys can be dark places even in daylight, thanks to the shadows of the tall buildings that surround them. A dark alley in daylight is rarely dark enough to afford true concealment, but it can lend a +2 circumstance bonus on Stealth checks.